

CAPE WRATH LH, United Kingdom

WMO: 030490

Lat: 58.6253N

Lon: 4.9995W

Elev: 112

StdP: 99.99

Time Zone: 0.00 (EUW)

Period: 86-97

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.4	0.5	-5.7	2.4	2.5	-4.6	2.6	2.7	29.7	5.0	26.3	6.6	7.7	140	0.925

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	19.4	14.8	17.6	14.2	16.2	13.6	15.7	18.0	14.8	16.8	13.9	15.6	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	10.6	16.8	13.8	10.0	15.7	13.1	9.5	14.5	44.3	18.2	41.8	17.0	39.3	15.7	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.8	21.3	19.3	-1.8	22.4	1.4	2.6	-2.8	24.3	-3.6	25.8	-4.4	27.2	-5.4	29.1		
(5)			-2.7	17.5	1.1	1.6	-3.5	18.7	-4.2	19.6	-4.8	20.5	-5.6	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	4.9	4.3	4.9	6.1	8.3	10.2	12.3	12.6	11.0	9.1	6.9	5.3		
	DBStd	3.63	2.17	2.06	2.20	1.95	2.74	2.29	2.31	2.17	1.84	2.24	2.06	2.27		
	HDD10.0	984	159	159	158	119	68	23	2	1	10	43	94	147		
	HDD18.3	3772	417	392	417	368	312	245	187	179	220	286	343	404		
	CDD10.0	256	0	0	0	1	15	28	74	81	40	16	1	0		
	CDD18.3	2	0	0	0	0	1	0	1	1	0	0	0	0		
	CDH23.3	2	0	0	0	0	0	0	0	2	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	8.5	11.0	10.4	9.8	7.4	6.9	6.3	6.6	6.9	8.2	8.8	9.3	10.2		
	PrecAvg	1370	156	124	121	88	82	72	79	98	114	138	147	152		
(15) Precipitation	PrecMax	1771	280	220	275	154	172	122	135	198	274	265	220	262		
(16) PrecipMin		1005	31	41	27	34	6	27	35	45	31	47	33	28		
(17) PrecStd		178	75	54	59	24	36	25	28	38	52	54	50	61		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	10.5	10.2	11.3	13.1	20.1	19.9	21.4	22.8	17.7	16.9	12.4	11.5			
	0.4% MCWB	8.9	8.3	8.2	9.6	13.6	13.9	16.0	17.5	14.5	13.9	10.2	9.6			
	2% DB	9.7	8.9	9.6	11.1	17.1	16.9	19.1	19.0	15.7	14.5	11.2	10.1			
	2% MCWB	8.2	7.7	7.6	9.1	12.3	13.0	15.2	15.4	13.2	12.0	9.9	9.2			
	5% DB	8.9	8.0	8.5	9.7	13.8	15.0	17.5	16.9	14.4	13.0	10.6	9.5			
	5% MCWB	7.9	7.2	7.0	8.1	10.3	12.5	14.5	14.4	12.5	11.2	9.4	8.7			
	10% DB	8.1	7.2	7.7	8.8	11.6	13.1	15.9	15.4	13.4	11.8	9.8	8.8			
	10% MCWB	7.2	6.4	6.6	7.6	9.3	11.3	13.7	13.7	12.0	10.2	8.7	7.9			
	0.4% WB	9.5	9.0	9.1	10.6	14.2	14.8	17.2	17.8	15.4	14.0	11.4	10.1			
	0.4% MCDB	10.0	9.7	10.0	12.2	18.8	17.6	20.3	21.8	16.6	16.0	11.8	10.8			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	8.8	8.1	8.1	9.4	12.4	13.5	15.6	16.0	13.8	12.5	10.5	9.5			
	2% MCDB	9.3	8.5	8.9	10.5	16.7	16.1	18.0	18.2	14.9	14.1	10.9	10.0			
	5% WB	8.1	7.4	7.5	8.5	10.8	12.5	14.7	15.0	13.0	11.5	9.8	8.8			
	5% MCDB	8.6	7.8	8.2	9.3	13.0	14.6	17.0	16.6	13.8	12.5	10.2	9.3			
	10% WB	7.4	6.6	6.9	7.8	9.6	11.5	13.9	13.8	12.3	10.7	8.9	8.1			
	10% MCDB	8.1	7.2	7.5	8.6	11.0	13.0	15.7	15.0	13.2	11.5	9.5	8.7			
(35) Mean Daily Temperature Range	MDBR	2.8	2.7	2.9	2.9	3.0	2.8	2.8	2.8	2.6	2.7	2.6	2.7			
	5% DB MCDBR	3.6	3.3	4.0	4.2	6.5	5.9	5.4	5.4	4.1	4.6	3.5	3.4			
	5% DB MCWBR	3.8	3.5	3.3	3.2	3.8	3.3	3.3	3.3	3.3	3.8	3.9	3.8			
	5% WB MCDBR	3.3	3.2	3.6	3.7	5.8	5.2	4.9	5.1	3.3	4.0	3.2	3.3			
	5% WB MCWBR	3.7	3.7	3.4	3.2	3.8	3.2	3.3	3.5	3.3	3.8	3.9	3.8			
(40) Clear-Sky Solar Irradiance	taub	0.326	0.345	0.360	0.375	0.372	0.366	0.372	0.362	0.355	0.340	0.320	0.307			
	taud	2.148	2.263	2.324	2.331	2.393	2.448	2.445	2.480	2.474	2.385	2.206	2.095			
	Ebn at Noon	464	665	777	830	860	871	855	839	783	670	472	351			
	Edh at Noon	51	74	91	106	107	103	101	91	79	64	47	38			
	RadAvg	0.32	0.90	1.98	3.48	4.70	4.65	4.31	3.52	2.32	1.18	0.44	0.21			
(44) All-Sky Solar Radiation	RadStd	0.04	0.11	0.14	0.23	0.33	0.41	0.31	0.20	0.15	0.10	0.05	0.02			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page